Database Technologies – Assignment 11 **KD2-Abhishek-96835**

1) Create a view that shows all of the customers who have the highest ratings.

```
KD2_Abhishek_96835>CREATE VIEW v_cust AS SELECT * FROM customers WHERE rating=(SELECT MAX(rating) from customers);
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from v_cust;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
2 rows in set (0.02 sec)
```

2) Create a view that shows the number of salespeople in each city.

```
KD2_Abhishek_96835>create view v_no_sp as select COUNT(snum) as count_of_salespeople, city from salespeople group by city;
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from v_no_sp;
+----------------------+-----------+
| count_of_salespeople | city      |
+----------------------+-----------+
|                    2 | London    |
|                    1 | San Jose  |
|                    1 | Barcelona |
|                    1 | New York  |
+----------------------+-----------+
4 rows in set (0.01 sec)
```

3) Create a view that shows the average and total orders for each salesperson after his or her name. Assume all names are unique.

```
KD2_Abhishek_96835>create view v_salesp_ords as select s.sname,AVG(o.amt) av
erage,count(o.onum) total from orders o inner join salespeople s on o.snum=s
.snum group by s.sname;
Query OK, 0 rows affected (0.02 sec)

KD2_Abhishek_96835>select * from v_salesp_ords;
+---------+-------------+-------+
| sname   | average     | total |
+---------+-------------+-------+
| Rifkin  |  558.425000 |     2 |
| Peel    | 5127.356667 |     3 |
| Motika  | 1900.100000 |     1 |
| Serres  | 1848.716667 |     3 |
| Axelrod | 1713.230000 |     1 |
+---------+-------------+-------+
5 rows in set (0.00 sec)
```

4) Create a view that shows each salesperson with multiple customers. Changing Values through Views.

```
KD2_Abhishek_96835>create view v_salesp_custs as select snum,sname from salespeople where snum in(select snum from customers group by snum having COUNT(cnum
)>1);
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from v_salesp_custs;
+------+--------+
| snum | sname  |
+------+--------+
| 1001 | Peel   |
| 1002 | Serres |
+------+--------+
2 rows in set (0.00 sec)
```

5) Which of these views are updateable (will allow DML operations)?

#1 Create View Dailyorders as Select Distinct cnum, snum, onum, odate from Orders;

```
KD2_Abhishek_96835>Create View Dailyorders
    -> as Select Distinct cnum, snum, onum, odate from Orders;
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from Dailyorders;
+------+------+------+------------+
| cnum | snum | onum | odate      |
+------+------+------+------------+
| 2008 | 1007 | 3001 | 1990-10-03 |
| 2001 | 1001 | 3003 | 1990-10-03 |
| 2007 | 1004 | 3002 | 1990-10-03 |
| 2003 | 1002 | 3005 | 1990-10-03 |
| 2008 | 1007 | 3006 | 1990-10-03 |
| 2002 | 1003 | 3009 | 1990-10-04 |
| 2004 | 1002 | 3007 | 1990-10-04 |
| 2006 | 1001 | 3008 | 1990-10-04 |
| 2004 | 1002 | 3010 | 1990-10-04 |
| 2006 | 1001 | 3011 | 1990-10-04 |
+------+------+------+------------+
10 rows in set (0.00 sec)

KD2_Abhishek_96835>delete from Dailyorders where cnum=2008;
ERROR 1288 (HY000): The target table Dailyorders of the DELETE is not updatable
```

#2 Create View Custotals as Select cname, Sum (amt) Sum_Amt from Orders, Customers where Orders.cnum=Customers.cnum Group by cname;

```
KD2_Abhishek_96835>Create View Custotals
    -> as Select cname, sum(amt) Sum_Amt from Orders, Customers
    -> where Orders.cnum=Customers.cnum
    -> Group by cname;
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from Custotals;
+----------+----------+
| cname    | Sum_Amt  |
+----------+----------+
| Cisneros |  1116.85 |
| Hoffman  |   767.19 |
| Pereira  |  1900.10 |
| Liu      |  5160.45 |
| Giovanni |  1713.23 |
| Grass    |   385.70 |
| Clemens  | 14614.88 |
+----------+----------+
7 rows in set (0.00 sec)

KD2_Abhishek_96835>delete from Custotals where cname='Liu';
ERROR 1288 (HY000): The target table Custotals of the DELETE is not updatable
```

#3 Create view Thirdorders as Select * from Dailyorders where odate=’1990-10-03’;

```
KD2_Abhishek_96835>Create view Thirdorders
    -> as Select * from Dailyorders where
    -> odate='1990-10-03';
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from Thirdorders;
+------+------+------+------------+
| cnum | snum | onum | odate      |
+------+------+------+------------+
| 2008 | 1007 | 3001 | 1990-10-03 |
| 2001 | 1001 | 3003 | 1990-10-03 |
| 2007 | 1004 | 3002 | 1990-10-03 |
| 2003 | 1002 | 3005 | 1990-10-03 |
| 2008 | 1007 | 3006 | 1990-10-03 |
+------+------+------+------------+
5 rows in set (0.01 sec)

KD2_Abhishek_96835>insert into Thirdorders values(2009,1000,3009,'2026-05-06');
ERROR 1471 (HY000): The target table Thirdorders of the INSERT is not insertable-into
KD2_Abhishek_96835>insert into Thirdorders (cnum) values(2009);
ERROR 1471 (HY000): The target table Thirdorders of the INSERT is not insertable-into
KD2_Abhishek_96835>
```

#4 Create view Nullcities as Select snum, sname, city from Salespeople where city is NULL OR sname BETWEEN ‘A’ and ‘MZ’;

```
KD2_Abhishek_96835>Create view Nullcities
    -> as Select snum, sname, city
    -> from Salespeople
    -> where city is NULL
    -> OR sname BETWEEN 'A' and 'MZ';
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from Nullcities;
+------+---------+----------+
| snum | sname   | city     |
+------+---------+----------+
| 1004 | Motika  | London   |
| 1003 | Axelrod | New York |
+------+---------+----------+
2 rows in set (0.00 sec)

KD2_Abhishek_96835>START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

KD2_Abhishek_96835>update Nullcities set sname='Ok' where snum=1004;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

6) Create a view of the Salespeople table called Commissions. This view will include only the snum and comm fields. Through this view, someone could enter or change commissions, but only to values between .10 and .20.

```
KD2_Abhishek_96835>create view Commissions as select snum,comm from salespeople where comm between 0.10AND 0.20 WITH CHECK OPTION;
Query OK, 0 rows affected (0.01 sec)

KD2_Abhishek_96835>select * from Commissions;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.12 |
| 1002 | 0.13 |
| 1004 | 0.11 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
5 rows in set (0.00 sec)

KD2_Abhishek_96835>insert into Commissions values(1009,0.25);
ERROR 1369 (HY000): CHECK OPTION failed 'sales.commissions'
```